Summary

Lagrange interpolation is a versatile tool in computer science, facilitating data estimation and function approximation across various domains. While it offers simplicity and precision for small datasets, considerations like computational efficiency and potential overfitting are crucial when applying it to larger or more complex problems.

	(x_0)	(x_1)	(x_2)	(x_3)
x :	1	3	5	7
y :	24	120	336	720
	(y_0)	(y_1)	(y_2)	(y_3)

By Lagrange's interpolation formula, we have

$$y = f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}y_3$$

$$= \frac{(x-3)(x-5)(x-7)}{(1-3)(1-5)(1-7)}(24) + \frac{(x-1)(x-5)(x-7)}{(3-1)(3-5)(3-7)}(120)$$

$$+ \frac{(x-1)(x-3)(x-7)}{(5-1)(5-3)(5-7)}(336) + \frac{(x-1)(x-3)(x-5)}{(7-1)(7-3)(7-5)}(720)$$

$$= -\frac{1}{2}(x-3)(x-5)(x-7) + \frac{15}{2}(x-1)(x-5)(x-7)$$

$$-21(x-1)(x-3)(x-7) + 15(x-1)(x-3)(x-5)$$

$$= -\frac{1}{2}[x^3 - 15x^2 + 71x - 105] + \frac{15}{2}[x^3 - 13x^2 + 47x - 35]$$

$$-21[x^3 - 11x^2 + 31x - 21] + 15[x^3 - 9x^2 + 23x - 15]$$

$$= \left[-\frac{1}{2} + \frac{15}{2} - 21 + 15\right]x^3 + \left[\frac{15}{2} - \frac{195}{2} + 231 - 135\right]x^2$$

$$+ \left[-\frac{71}{2} + \frac{705}{2} - 605 + 345\right]x + \left[\frac{105}{2} - \frac{525}{2} + 441 - 225\right]$$

$$= x^3 + 6x^2 + 11x + 6$$

$$f(4) = 4^3 + 6(4^2) + 11(4) + 6 = 64 + 96 + 44 + 6 = 210$$

Note : (1) $(x-a)(x-b) = x^2 - (a+b)x + ab$

(2) $(x-a)(x-b)(x-c) = x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc$

4. Using Lagrange interpolation find y (2) from the following data.

x :	0	1	3	4	5
y :	0	1	81	256	625

Solution: Given:

	(x_0)	(x_1)	(x_2)	(x_3)	(x_4)
x :	0	1	3	4	5
y :	0	1	81	256	625
	(y_0)	(y_1)	(y_2)	(y_3)	(y_4)

By Lagrange's interpolation formula, we have

$$y = f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} y_3 + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)} y_4$$

$$y(2) = \frac{(2-1)(2-3)(2-4)(2-5)}{(0-1)(0-3)(0-4)(0-5)}(0) + \frac{(2-0)(2-3)(2-4)(2-5)}{(1-0)(1-3)(1-4)(1-5)}(1) + \frac{(2-0)(2-1)(2-4)(2-5)}{(3-0)(3-1)(3-4)(3-5)}(81) + \frac{(2-0)(2-1)(2-3)(2-5)}{(4-0)(4-1)(4-3)(4-5)}(256) + \frac{(2-0)(2-1)(2-3)(2-4)}{(5-0)(5-1)(5-3)(5-4)}(625)$$

$$= \frac{(2)(-1)(-2)(-3)}{(1)(-2)(-3)(-4)} + \frac{(2)(1)(-2)(-3)}{(3)(2)(-1)(-2)}(81) + \frac{(2)(1)(-1)(-3)}{(4)(3)(1)(-1)}(256) + \frac{(2)(1)(-1)(-2)}{(5)(4)(2)(1)}(625)$$

$$= \frac{12}{24} + \frac{12}{12}(81) - \frac{6}{12}(256) + \frac{4}{40}(625) = \frac{1}{2} + 81 - 128 + 62.5$$

$$= 0.5 + 81 - 128 + 62.5 = 16$$

5. Using Lagrange's interpolation formula, find f(4) given that f(0) 2, f(1) = 3, f(2) = 12, f(15) = 3587

Solution: Given

	(x_0)	(x_1)	(x_2)	(x_3)
x	0	1	2	15
y	2 (y_0)	3 (y_1)	12 (y_2)	3587 (y_3)

By Lagrange's interpolation formula, we have

$$y = f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}y_3$$

$$y = f(4) = \frac{(4-1)(4-2)(4-15)}{(0-1)(0-2)(0-15)}(2) + \frac{(4-0)(4-2)(4-15)}{(1-0)(1-2)(1-15)}(3) + \frac{(4-0)(4-1)(4-15)}{(2-0)(2-1)(2-15)}(12) + \frac{(4-0)(4-1)(4-2)}{(15-0)(15-1)(15-2)}(3587)$$

$$= \frac{(3)(2)(-11)}{(-1)(-2)(-15)}(2) + \frac{(4)(2)(-11)}{(1)(-1)(-14)}(3) + \frac{(4)(3)(-11)}{(2)(1)(-13)}(12) + \frac{(4)(3)(2)}{(15)(14)(13)}(3587)$$

$$= \frac{132}{30} - \frac{264}{14} + \frac{1584}{26} + \frac{86088}{2730} = 78$$

6. Using Lagrange's interpolation formula, find y(10) given that y(5) = 12, y(6) = 13, y(9) = 14, y(11) = 16.

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Solution:

$$y = f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)}y_0 + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)}y_1 + \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)}y_2 + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)}y_3$$

$$= \frac{(x-6)(x-9)(x-11)}{(5-6)(5-9)(5-11)}(12) + \frac{(x-5)(x-9)(x-11)}{(6-5)(6-9)(6-11)}(13) + \frac{(x-5)(x-6)(x-11)}{(9-5)(9-6)(9-11)}(14) + \frac{(x-5)(x-6)(x-9)}{(11-5)(11-6)(11-9)}(16)$$

Putting $x = 10$

$$y(10) = f(10) = \frac{(4)(1)(-1)}{(-1)(-4)(-6)}(12) + \frac{(5)(1)(-1)}{(1)(-3)(-5)}(13) + \frac{(5)(4)(-1)}{(4)(3)(-2)}(14) + \frac{(5)(4)(1)}{(6)(5)(2)}(16)$$

$$y = 14.666666 \Rightarrow y = 15$$

7. Using Lagrange's fprmula, fit a polynomial to the data

x	-1	1	2
y	7	5	15